BCSC0007: INTRODUCTION TO MICROPROCESSORS

Objective: *Objective of this subject is to introduce the basic concepts of microprocessor and assembly language programming. Identify and explain the operation of the components of typical microprocessor: the role of the ALU, registers, stack and the use of interrupts.*

Credits: 03 **L–T–P-J: 3–0–0-0**

Module No.	Content	Teaching Hours
I	**Introduction**: Microprocessors Evolution and Types, Basics of Pentium Microprocessor, Microprocessor Application, **8-Bit Microprocessor:**8085 Microprocessor and its Architecture, Addressing Modes, The 8085 Programming Model, Instruction Classification, Instruction Format, Overview of Instruction Set - Data Transfer Operation, Arithmetic Operation, Logic Operations and Branch Operations; Introduction to Assembly Language Program.	13
II	**Programming Technique with Additional Instruction**: Looping, Counting, Indexing, Additional Data Transfer and 16-Bit Arithmetic Instruction, Counters and Time Delays, Stack and Subroutine. **16 Bit Microprocessor:** Architecture of 8086 – Register Organization, Execution Unit, Bus Interface Unit, Signal Description, Physical Memory Organization, Mode of Operation, I/O Addressing Capabilities.	14
III	**Peripheral Interfacing:** I/O Programming, Programmed I/O, Interrupt Driven I/O, DMA I/O, Memory-Mapped I/Os. **Peripheral Devices:** 8237 DMA Controller, 8255 Programmable Peripheral Interface, 8253/8254 Programmable Timer/Counter, 8259 Programmable Interrupt Controller.	13

Text Books:

- N Senthil Kumar, M Saravanan, and S Jeevananthan (2010), "*Microprocessors and Microcontrollers*", Oxford University Press India.

Reference Books:

- Ramesh S. Gaonkar (2000), "*Microprocessor Architecture Programming and Applications with 8085*", 4th Edition, Penram International Publishing.
- Ray A.K.Bhurchandi.K.M (2002), "*Advanced Microprocessor and Peripherals*", TMH.
- D. V. Hall (1992), "*Microprocessors and Interfacing: Programming and Hardware*", 2nd Edition, TMH.
- Y.C. Liu and G.A. Gibson (2003), "*Microcomputer Systems: The 8086/8088 Family Architecture Programming and Design*", 2nd Edition, PHI.

Outcome: After the completion of the course, the student will be able to:

- Understand the processor organization, behavior of processing elements in different addressing modes.
- Analyze the data and instructions flow inside the internal components like computing elements, memory and input output devices